EEE522 Autonomous Vehicles

Vehicle States and Dynamics

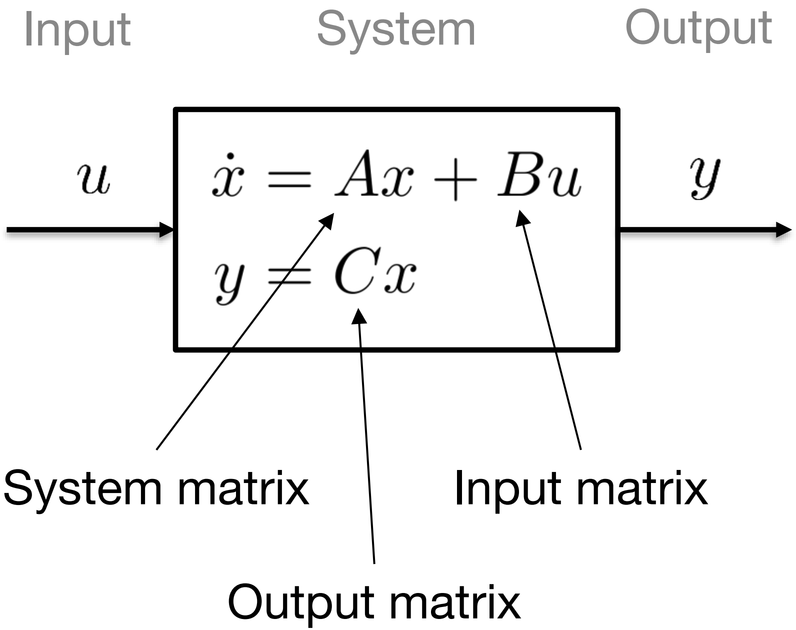
Ivan Ho

Outline

- Vehicle States
- Vehicle Dynamics Modeling
- Vehicle Dynamics Simulation

Vehicle States

System Dynamics



- A dynamic system is a mathematical description of the relation between an *input u* and an *output y* signal. This description is usually given in form of an ODE (Ordinary Differential Equation).
- The system has ***states x*****.** These are variables which allow us to **formulate the system behavior** in form of a set of first-order ODE for each of this variables.
- The standard system description consists of the ***system dynamics*** and the ***output equation***. The former describes the timely behavior of the states as a reaction to the inputs and the initial state. The latter describes the relation between the state and the output.

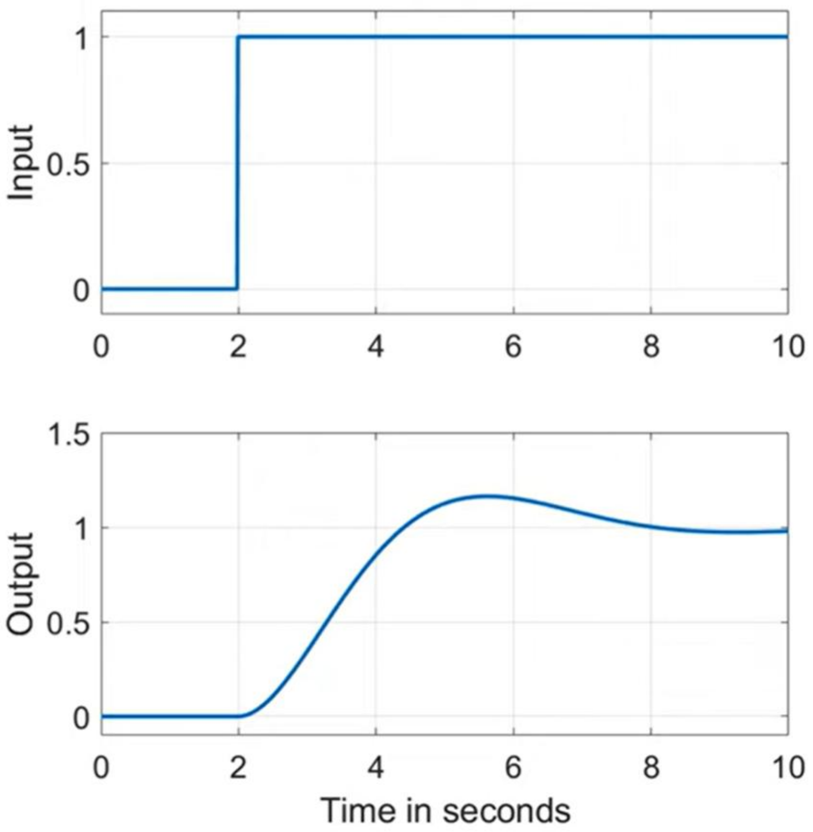
System Dynamics

Input System Output

$$u \rightarrow \boxed{\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned}} \rightarrow y$$

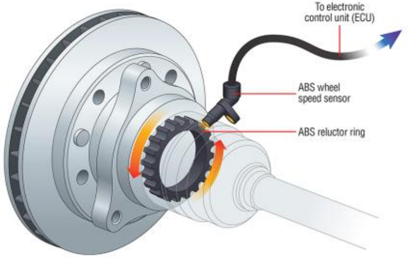
Input System Output

$$u \rightarrow \boxed{\begin{aligned} \dot{x} &= f(x, u) \\ y &= h(x) \end{aligned}} \rightarrow y$$

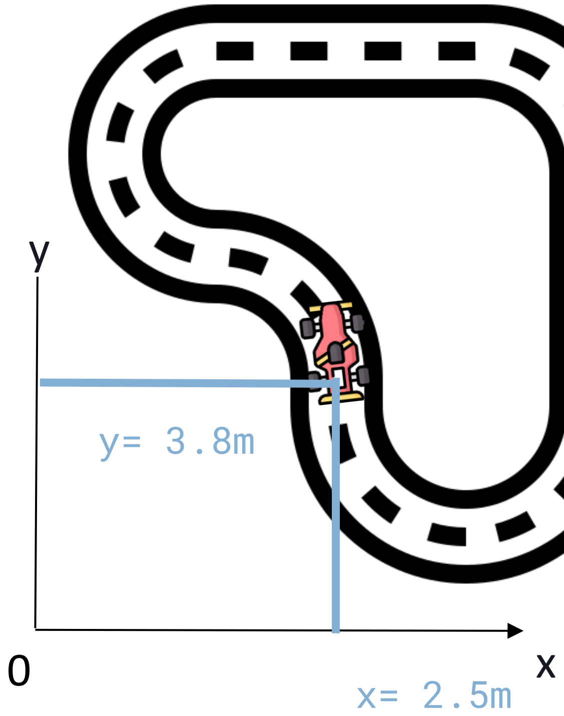


Measuring Vehicle States

- Vehicle states define the vehicle's behavior/position in its environment
- Vehicle states can be derived from different sensors
  - GPS (Global Positioning System)
  - Odometry: Gyrometer, Wheel Speed Sensor
  - Camera
  - LiDAR

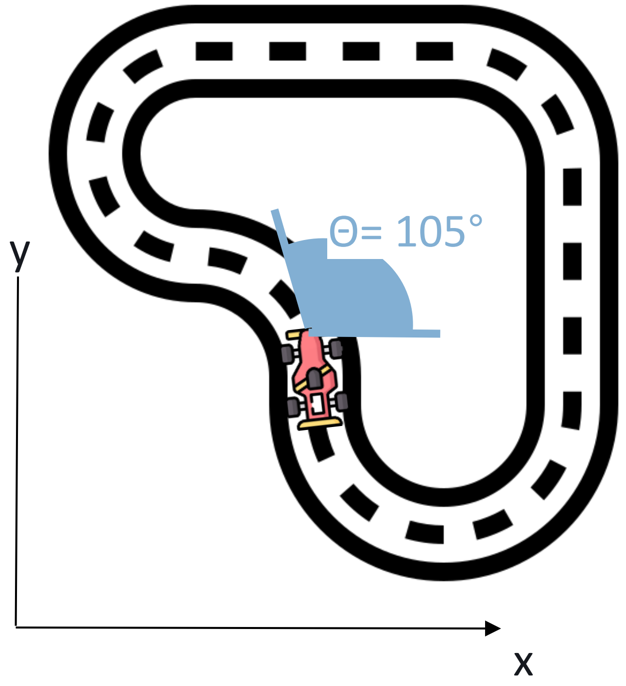


Position



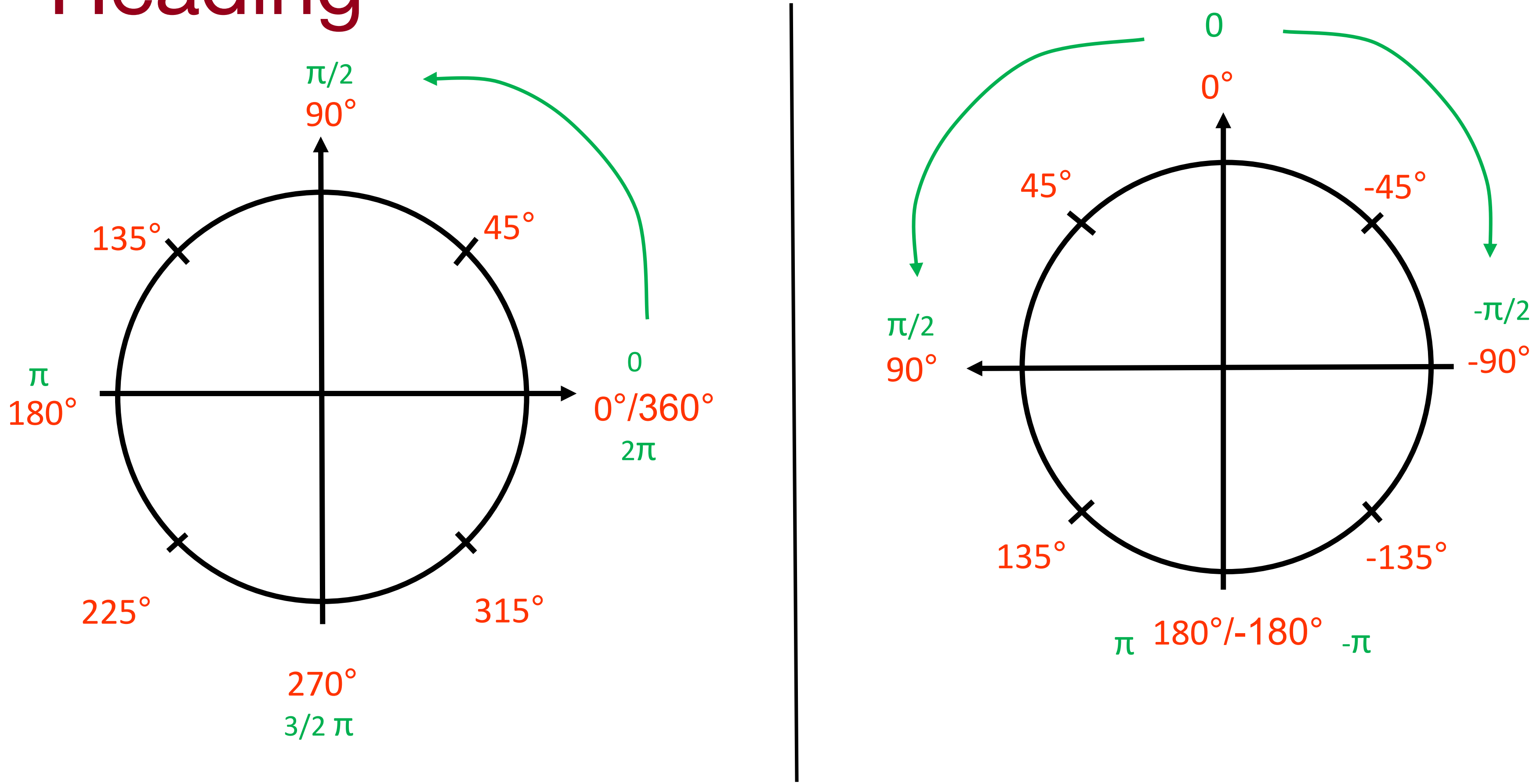
- **Position** defines the translation of the vehicle in some global or local frame
- Respective to the vehicle's center of gravity (CoG), or a pre-defined base frame
- **Normally:** X- and Y- position in **meters**

Heading



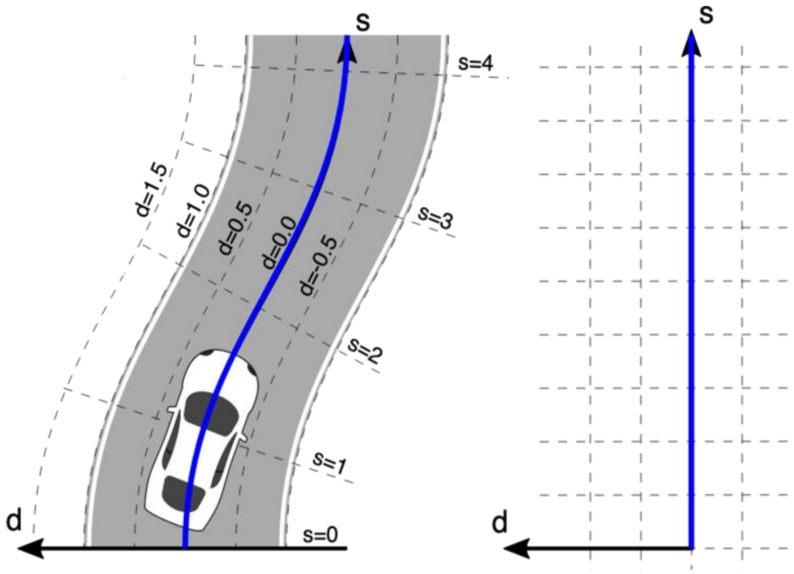
- **Heading** defines the rotation of the vehicle in some local and global frame
- Usually with respect to the x-axis of the coordinate system of current frame
- When represented as a single angle reading, heading can be displayed in ranges from:
  - $-\pi$ – π = -180° to 180°
  - 0 – 2π = 0° to 360°
- Only **rotation**

Heading



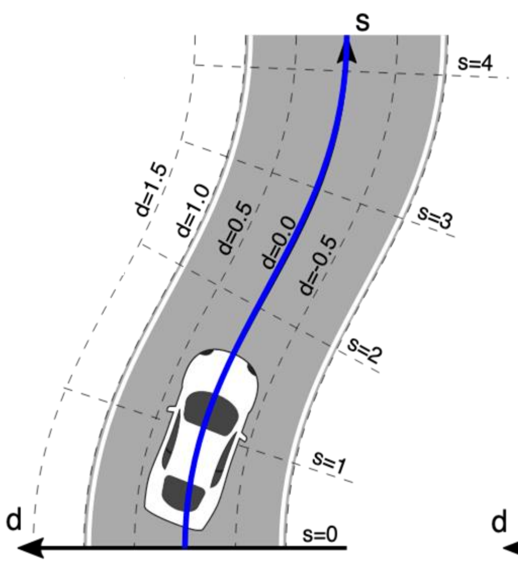
π/2
90°
45°
135°
π
180°
0
0°/360°
2π
225°
315°
270°
3/2 π
0
0°
45°
-45°
π/2
90°
-π/2
-90°
135°
-135°
π
180°/-180°
-π

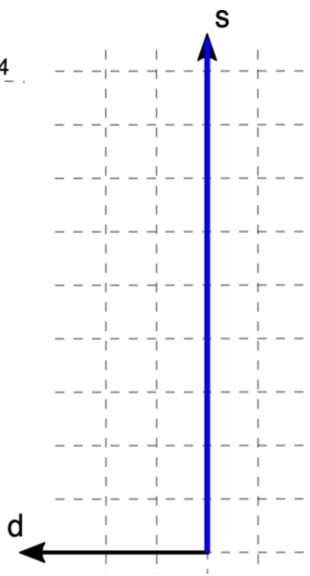
Position in Frenet Frame



- **The Frenet frame** along a curve is a moving coordinate system determined by the tangent line and curvature**.**
- The frame itself is defined as the coordinate system spanned by a **tangential vector T** and the **normal vector N**, and a **binormal vector B** at any point of the reference line (e.g., centerline of a track).
- (The binormal vector is the cross product of the tangential vector and the normal vector)

Position in Frenet Frame

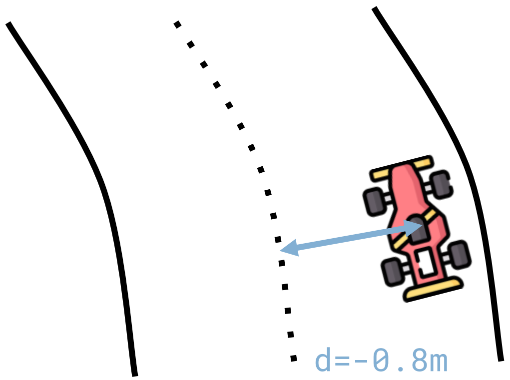




- Two coordinates in 2D.
- The **s** coordinate represents the run length. **Starts with s = 0** at the beginning of the reference path.
- The **d** coordinate represents lateral position relative to the reference path. **Starts with d = 0** for points on the reference path. Measured on the normal vector.
- **d is positive to the left** of the reference path and **negative to the right** of it.

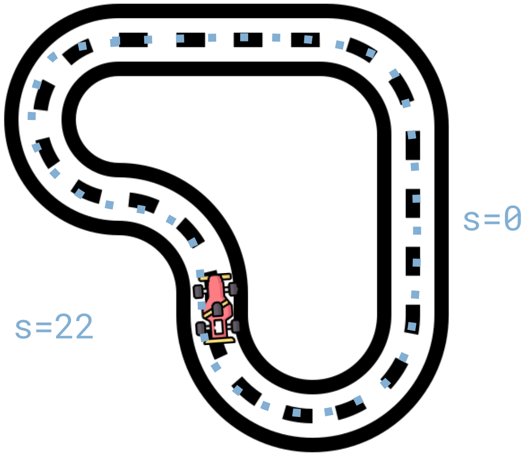
Frenet Frame in Practice

Distance to Centerline



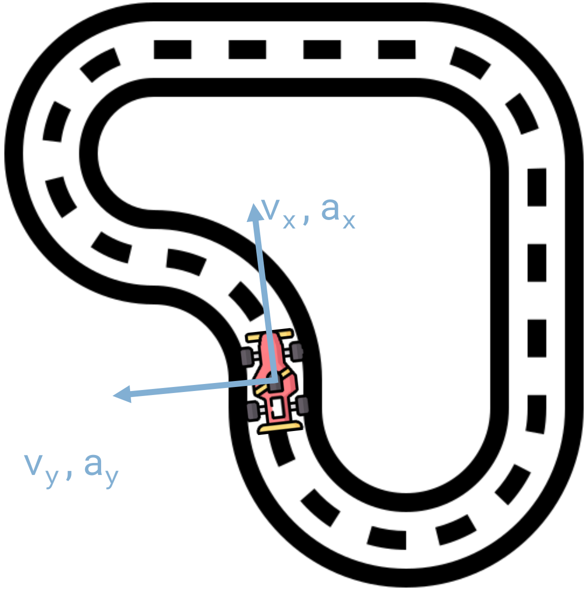
- Displacement (in meters) between the **agent center and the track center**
- The observable maximum displacement occurs when any of the agent's wheels are outside a track border and, depending on the width of the track border

Progress along the track



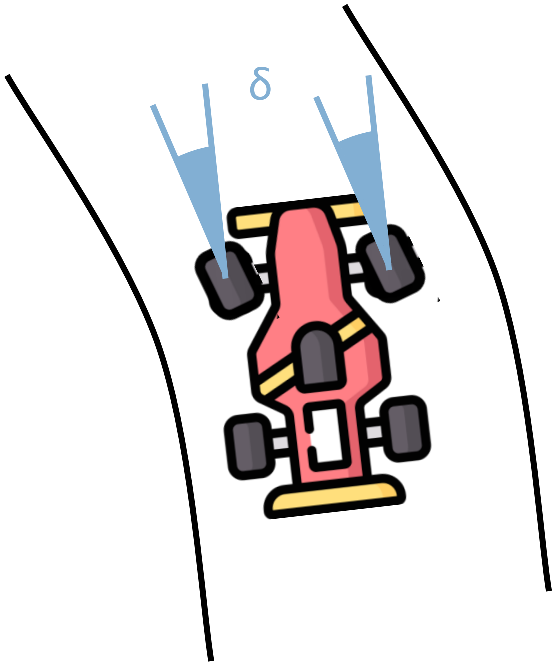
- An **ordered list** of track-dependent milestones along the track (waypoints).
- Each milestone is described by a coordinate of (x_i ; y_i, ...).
- For a looped track, **the first and last waypoints are the same.**

Linear Velocity and Acceleration



- **Linear velocity and acceleration** are measured in the x- and y-, (and z-) axis in the coordinate system of the vehicle.
- For cars: **longitudinal** (x-axis) and **lateral** (y-axis) **velocities** and **accelerations**.
- Can be measure with GPS, IMU, wheel speed sensors, pitot sensors, etc.
- Usually velocity in meters per second: m/s, and acceleration in meters per second squared: m/s^2

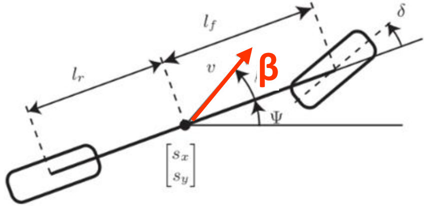
Vehicle State - Steering Angle



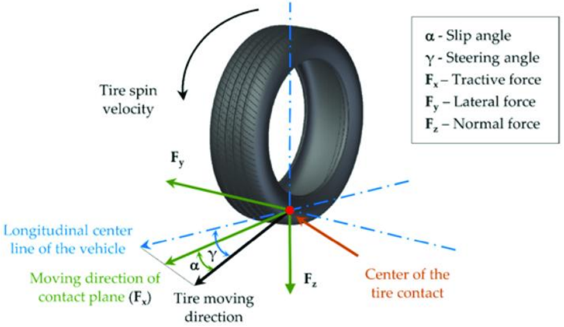
- **Steering angle δ** is the angle formed by the direction the front wheels "look at" and the vehicle's x-axis.
- It is defined w.r.t. the central line of the agent.
- Steering angle is the same for both front wheels.
- δ is usually displayed in radians or degrees.

Different Slip Angles

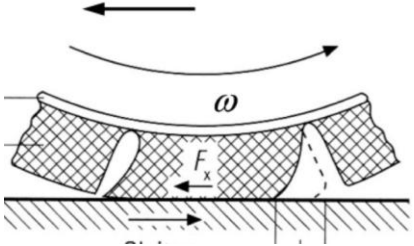
- **Sideslip angle *β***
  (when drifting)
  between direction of travel and x-axis of the chassis



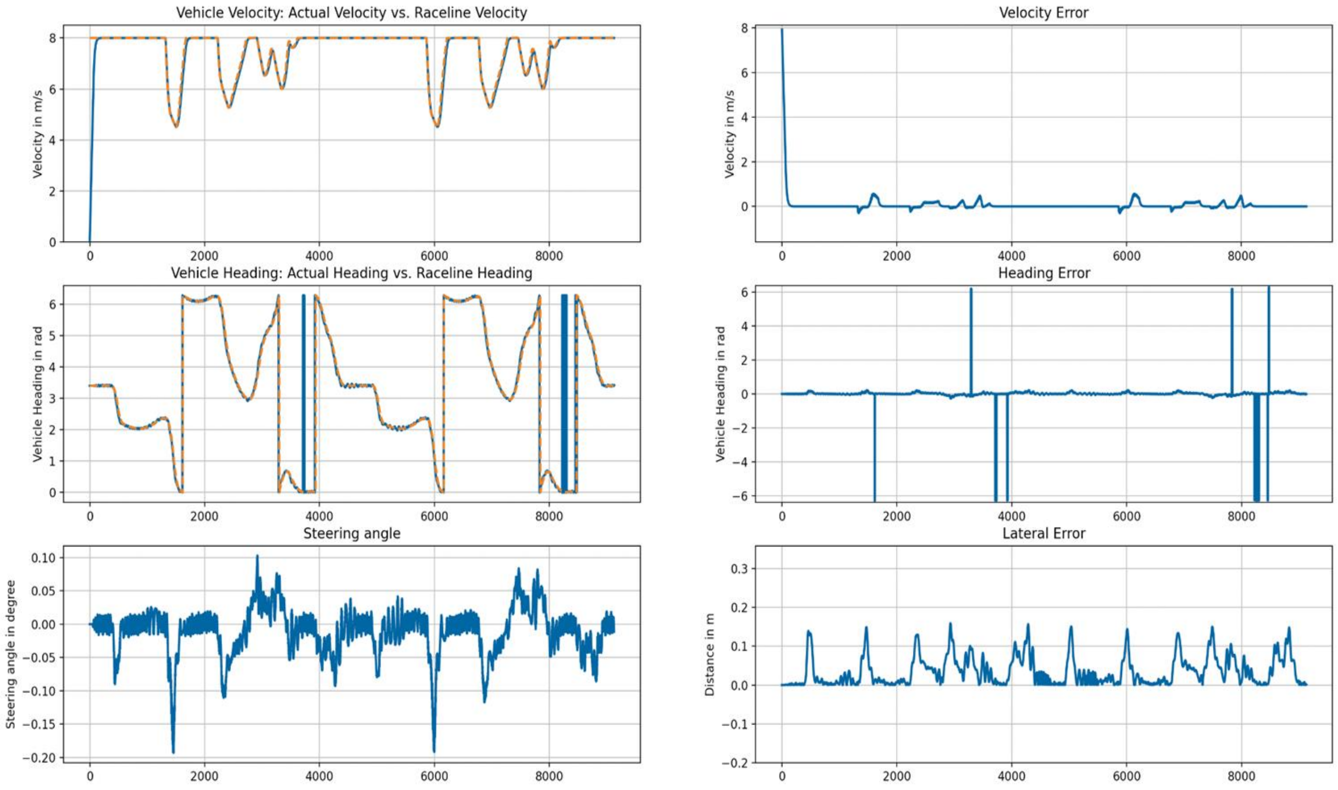
- **Slip Angle *α***
  (occurs at the tires when steering)
  between the direction of travel and the angle the wheel is pointing



- **Wheelslip ratio *s***
  (during a burnout)
  measure of how much a braked wheel is slipping longitudinally relative to pure rolling



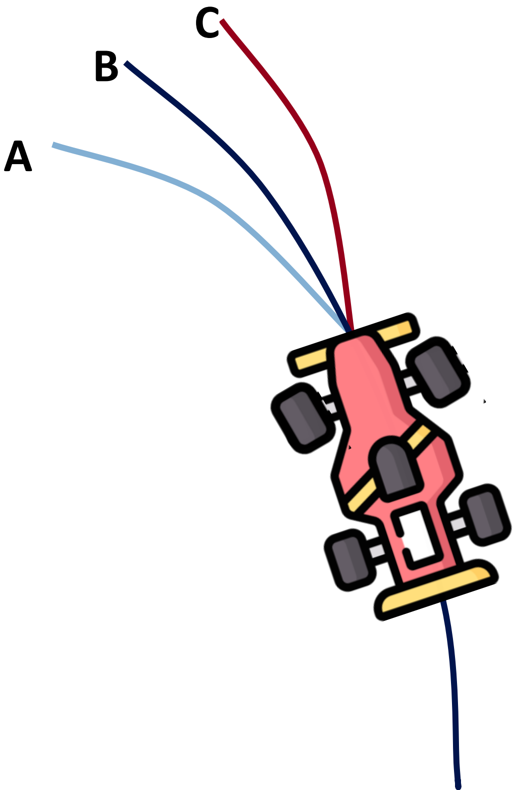
Important - Visualize your States



- Visualize
- Show trends
- Calculate errors
- Learn to Interpret the signals
- Plotjuggler, matplotlib, rqt_plot, etc.

Vehicle Dynamics Modeling

Why Modeling



Which trajectory is the car following?

We cannot tell

We need information about:

Velocity

Acceleration

Steering angle

Vehicle: Mass, Length, Width, ….

Why Modeling?

Inputs
(e.g. steering wheel
angle,motor torque)

Real Vehicle

Outputs
(e.g. speed,
accelerations)

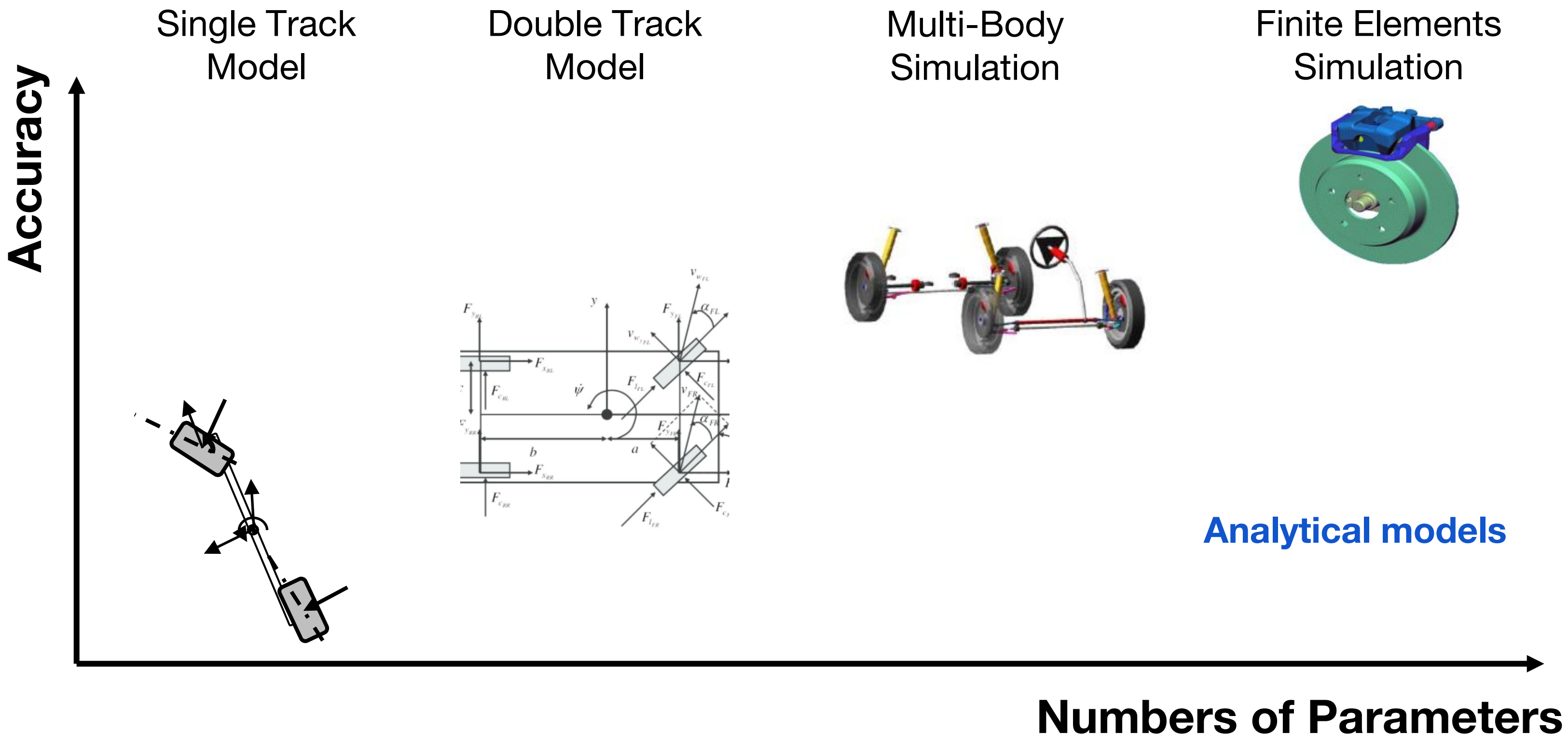
Simplified Representation

Model Accuracy and
Mismatch

Vehicle dynamics model

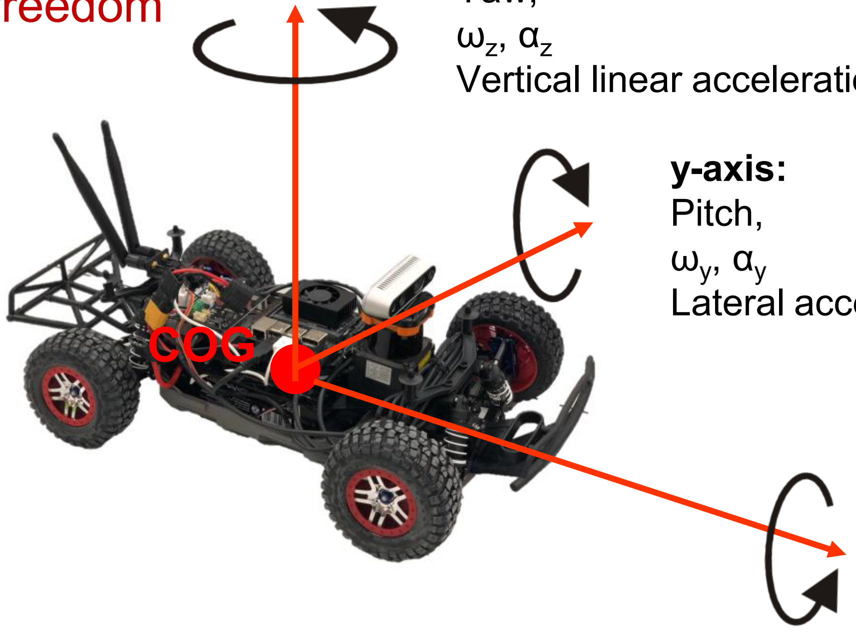
- Differential equations
- Characteristic curve correlations

Vehicle Dynamics Model Types



Angular Velocity and Acceleration

6 degrees of freedom



z-axis:
Yaw,
ω_z, α_z
Vertical linear acceleration (lift), v_z ,a_z

y-axis:
Pitch,
ω_y, α_y
Lateral acceleration v_y,a_y

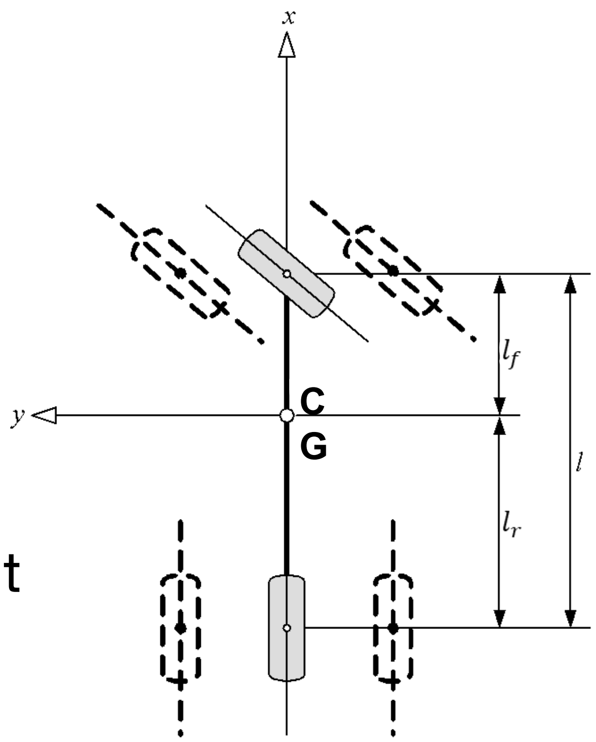
x-axis:
Roll,
ω_x, α_x
Longitudinal acceleration v_x,a_x

Single Track Models

Single Track Model

Simplifications:

- Wheels of one axle are combined (bicycle model)
- Center of gravity is at road level
  - No rolling
  - No pitching
  - No wheel load differences left/right
- Vehicle longitudinal speed is constant
- No vertical dynamics



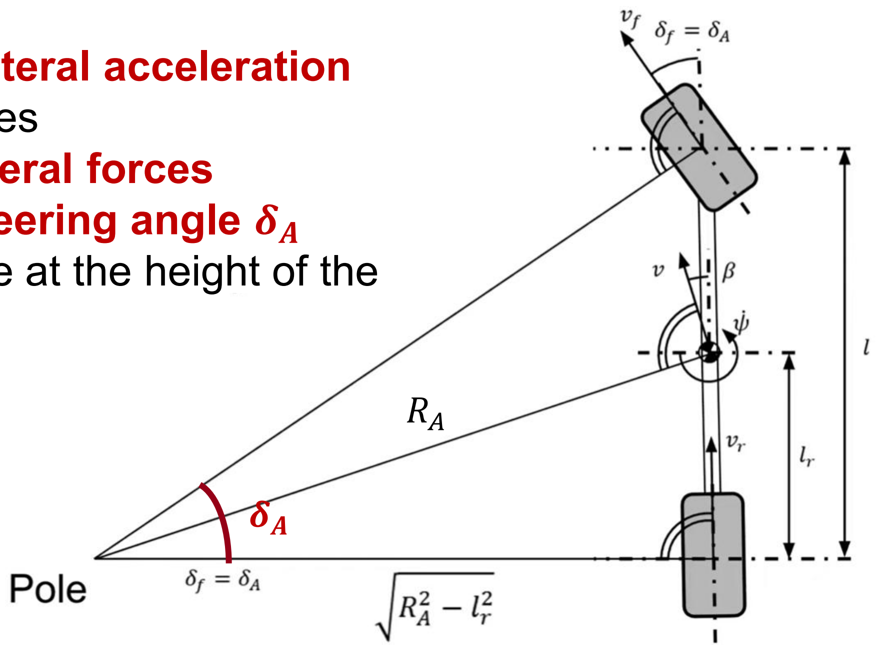
Kinematic Single Track Model

Kinematic:

- Slow driving, slow speed
- Cornering slowly → virtually **without lateral acceleration**
- The tires roll along their longitudinal axes
  - There are **no slip angles** and **no lateral forces**
  - Steering angle is the **Ackermann steering angle δ_A**
  - Driving around an instantaneous pole at the height of the rear axle.

$$\tan \delta_A = \frac{l}{\sqrt{R_A^2 - l_r^2}} \rightarrow \quad \delta_A \approx \frac{l}{R_A}$$

radius of curvature



Kinematic Single Track Model

$$\dot{x} = f(x, u)$$

$$x = [p_1 \quad p_2 \quad \psi]^{\mathrm{T}}$$

position

heading

$$u = [\delta \quad v]^{\mathrm{T}}$$

steering angle

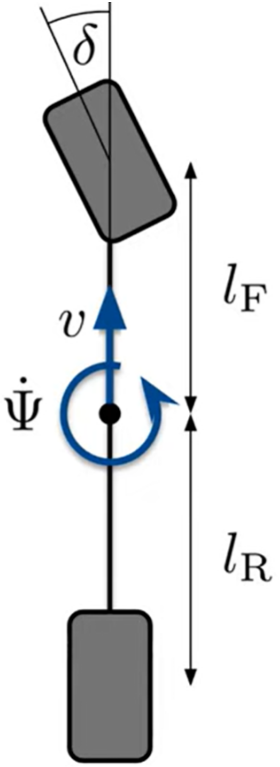
velocity

$$\dot{p}_1 = v \cos \Psi$$

$$\dot{p}_2 = v \sin \Psi$$

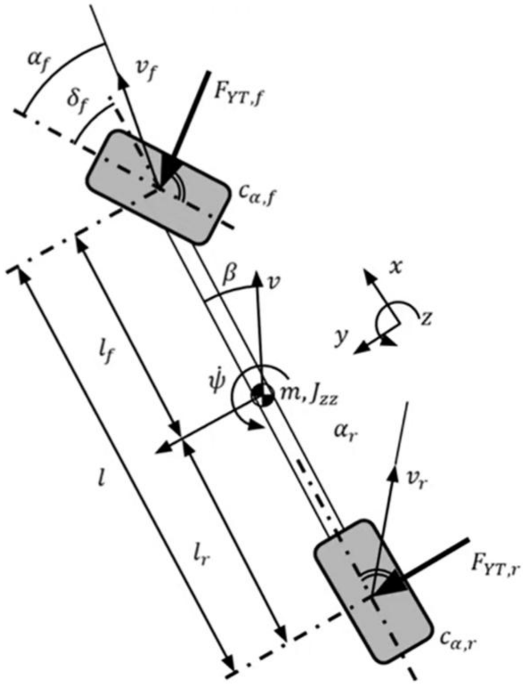
$$\dot{\Psi} = \frac{\tan(\delta)}{l_\mathrm{F} + l_\mathrm{R}} v$$

yaw rate



Linear Single Track Model

- We want to plan evasive maneuvers closer to physical limits
- Consider important effects such as understeer or oversteer
- **Steady-state velocity assumption** $\dot{v} = 0$
- **Introduction of tire forces**
    - The tire creates the contact between the vehicle and the road
    - A tire can apply **lateral and longitudinal tire forces**
    - A tire can apply more forces if there is a higher **friction coefficient μ**
    - **Linear relation** between **tire force** and **side slip angle β**
    - Model the tire dynamics with the **cornering stiffness c**



Linear Single Track Model

$$\dot{x} = f(x, u)$$

$$\dot{x} = Ax + Bu$$

$$x = [\beta \quad \dot{\psi}]^{\mathrm{T}}$$

chassis side slip

yaw rate

$$u = \delta$$

steering angle

$$A = \begin{bmatrix} -\frac{c_{\mathrm{F}}+c_{\mathrm{R}}}{mv} & -\frac{mv^2-(c_{\mathrm{R}}l_{\mathrm{R}}-c_{\mathrm{F}}l_{\mathrm{F}})}{mv^2} \\ \frac{c_{\mathrm{R}}l_{\mathrm{R}}-c_{\mathrm{F}}l_{\mathrm{F}}}{I_z} & -\frac{c_{\mathrm{R}}l_{\mathrm{R}}^2-c_{\mathrm{F}}l_{\mathrm{F}}^2}{I_z v} \end{bmatrix}$$

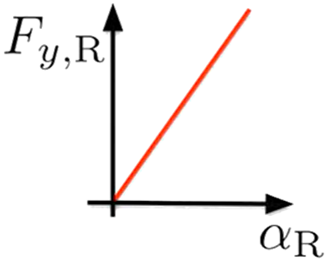
$$B = \begin{bmatrix} \frac{c_{\mathrm{F}}}{mv} \\ \frac{c_{\mathrm{F}}l_{\mathrm{F}}}{I_z} \end{bmatrix}$$

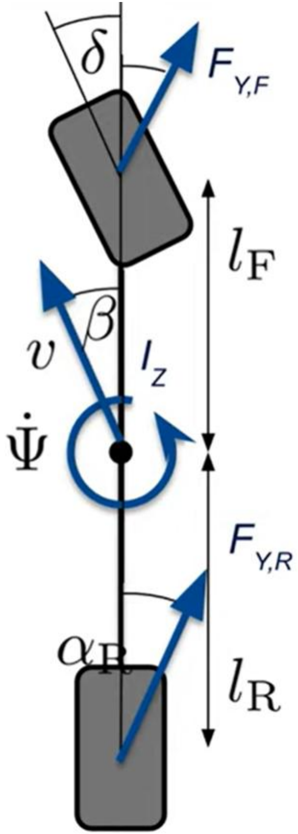
tire force

cornering stiffness

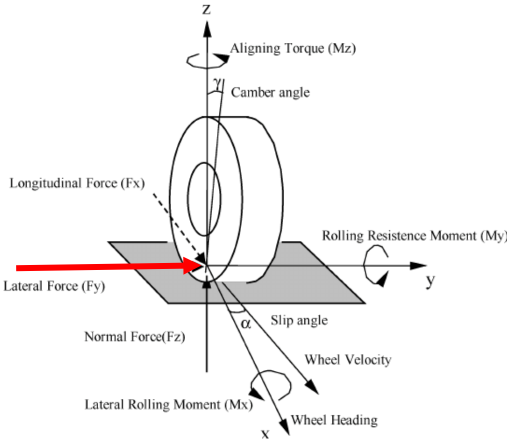
$$c_{\mathrm{R}} = \frac{\partial F_{y,\mathrm{R}}}{\partial \alpha_{\mathrm{R}}}$$

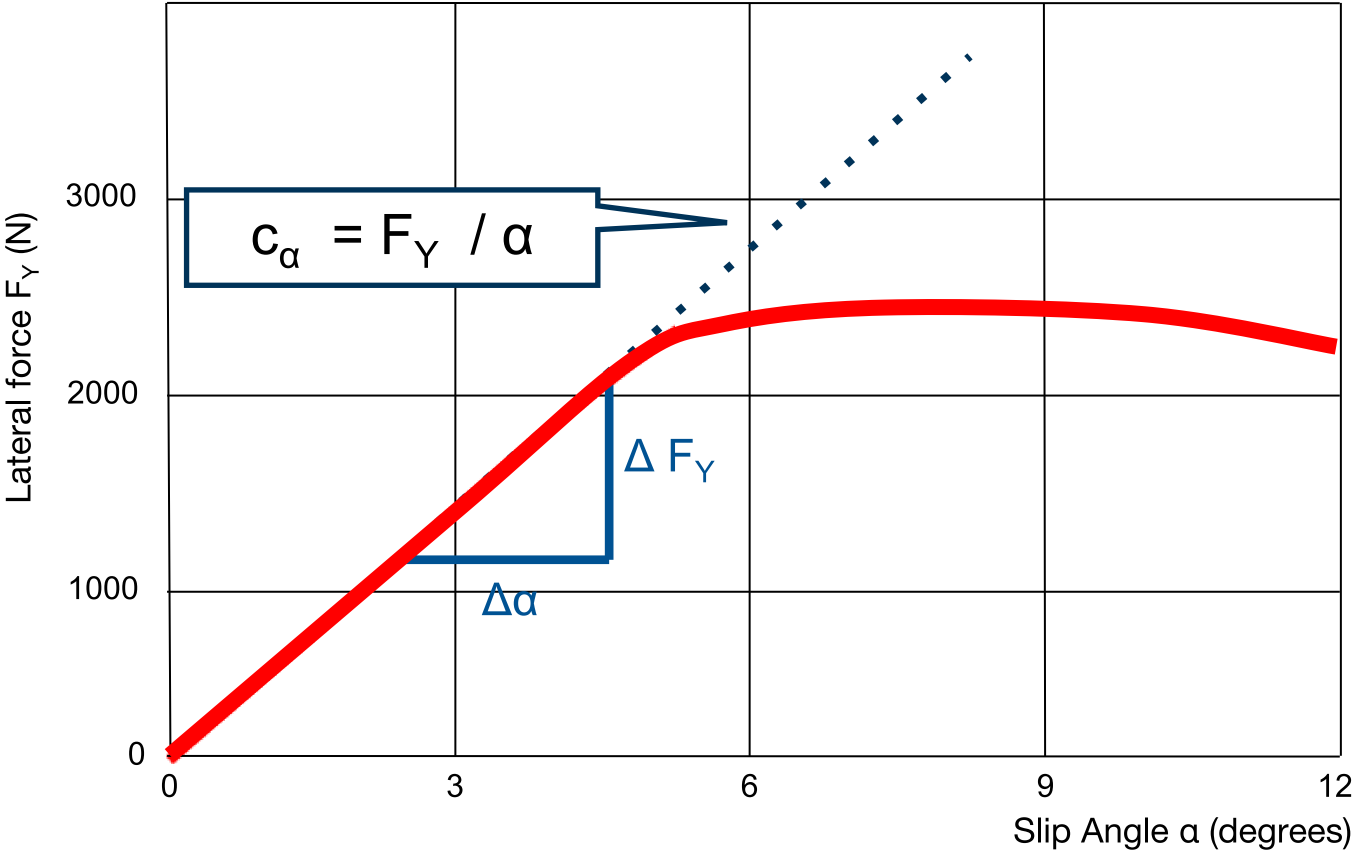
slip angle





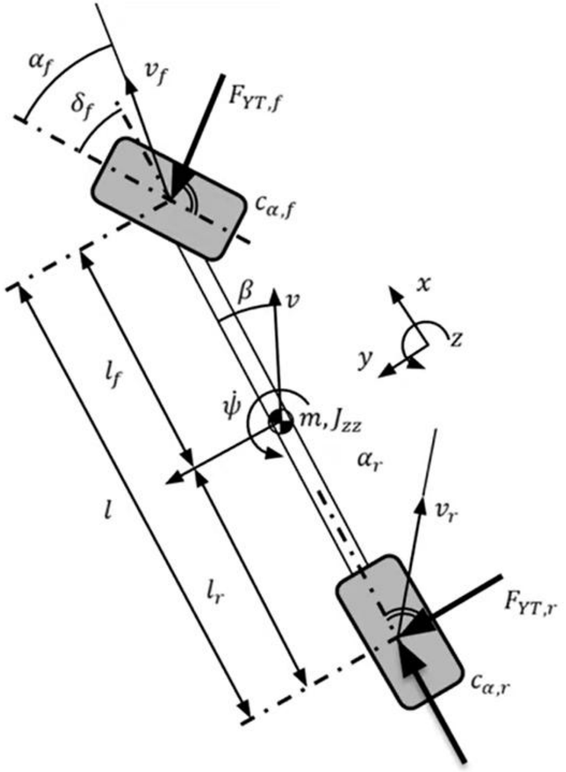
Linear Tire Model





Non-linear Single Track Model

- We want to plan evasive maneuvers closer to physical limits
- Consider important effects such as understeer or oversteer
- **Variable Velocity:**
  - Velocity is dependent on longitudinal forces
  - Velocity is dependent on longitudinal resistances
- **Extension of tire forces**
  - **Nonlinear relation** between **tire force** and **side slip angle**
  - Model the tire dynamics with the special models
    - Pacejka Magic Formula
    - Fiala



Non-linear Single Track Model

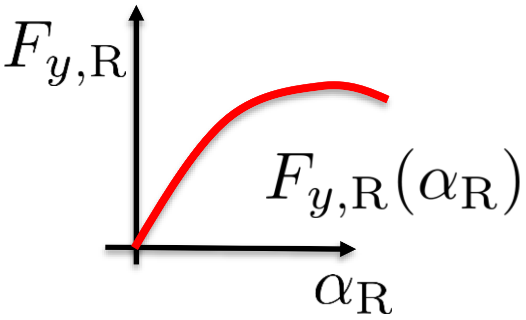
$$\dot{x} = f(x, u)$$

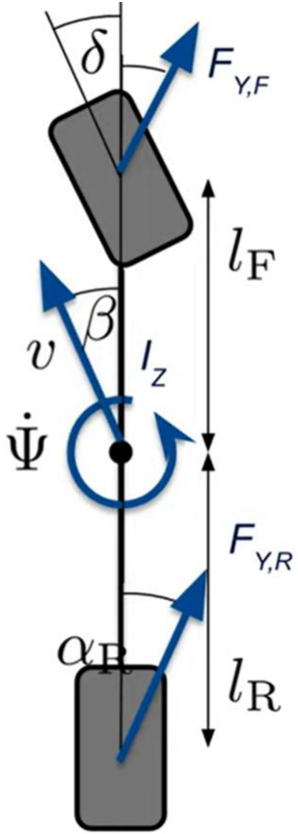
$$x = [p_1 \quad p_2 \quad \psi \quad v_x \quad v_y \quad \dot{\psi}]^{\mathrm{T}}$$

position heading long. velocity lat. velocity yaw rate

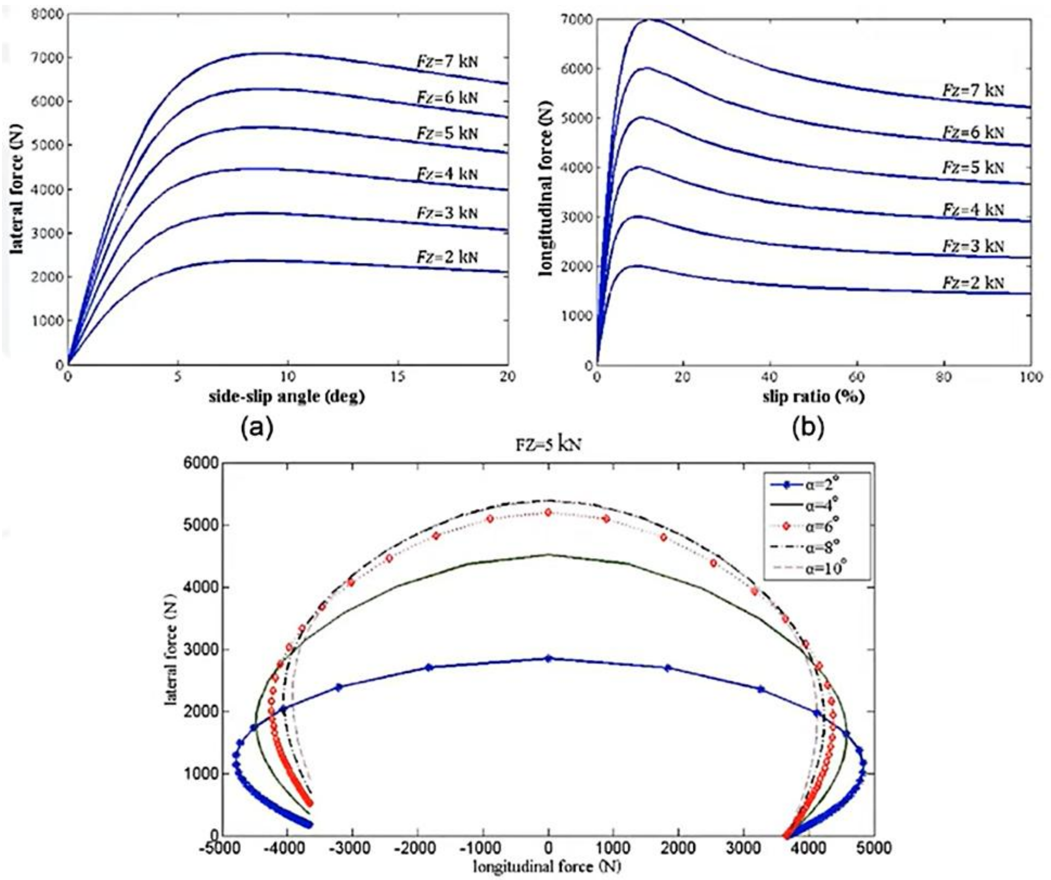
$$u = [\delta \quad \boldsymbol{F}]^{\mathrm{T}}$$

steering angle tire force

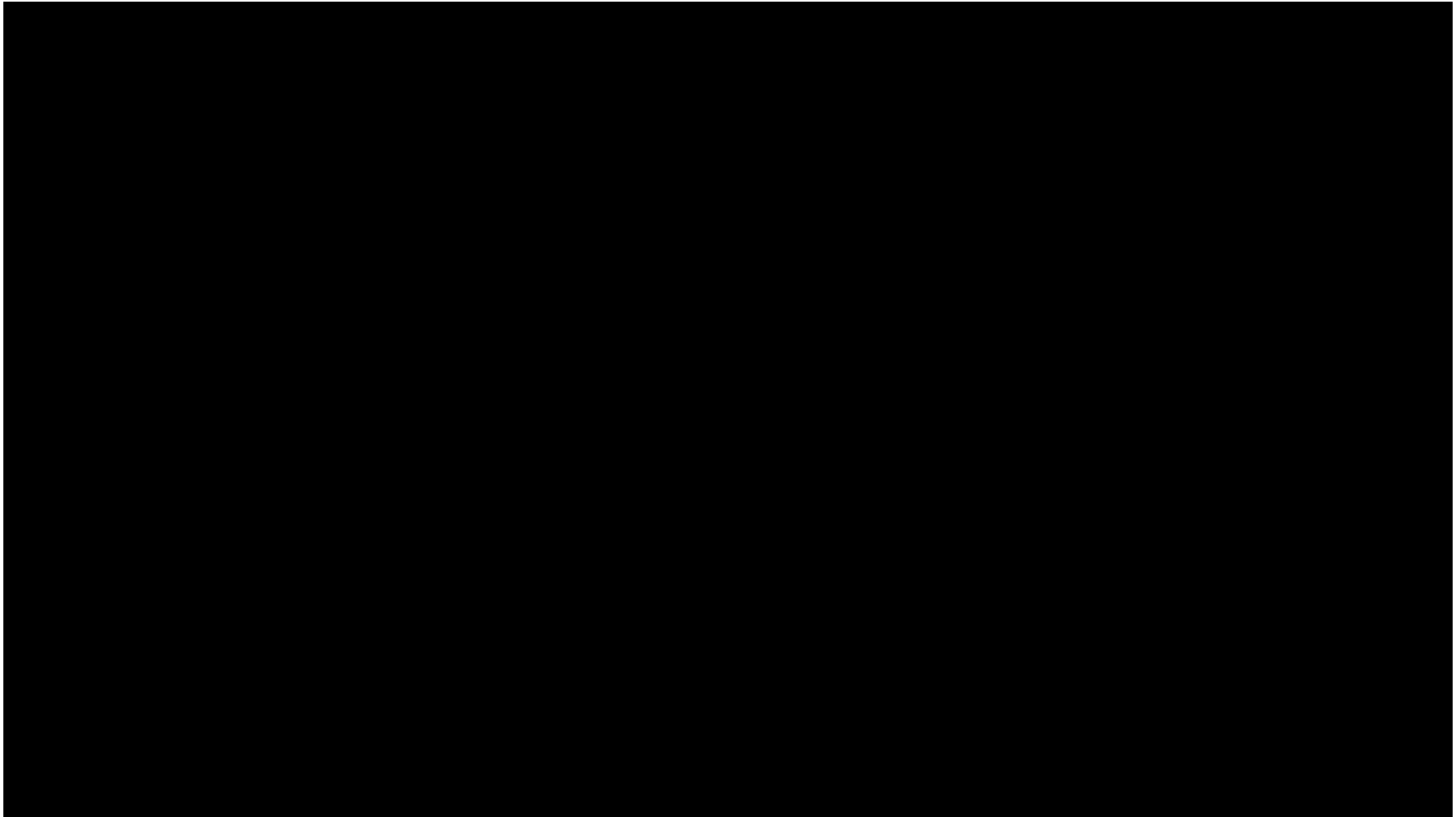




Non-linear Single Track Model



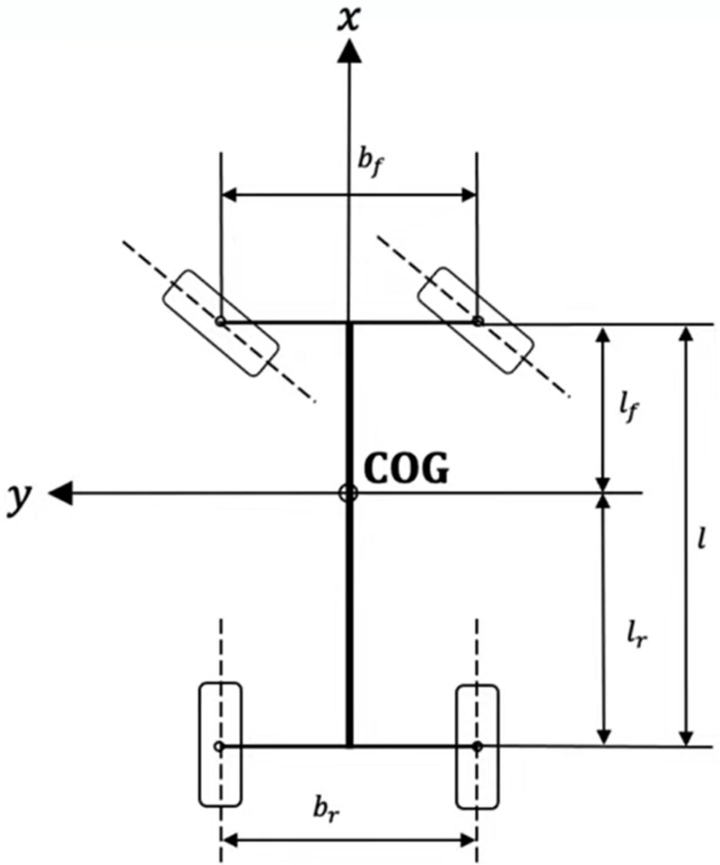
Exceeding the maximum longitudinal and lateral tire forces…



Double Track Model

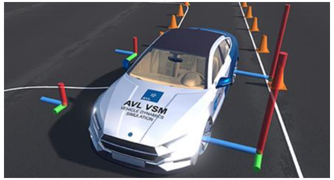
Complicated vehicle model, model all dynamics

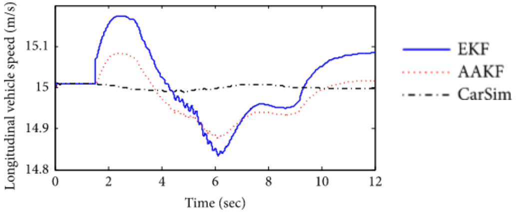
- Each wheel is modeled individually
- Each wheel has individual wheel load
- Each wheel has individual lat/long forces
- Center of gravity is at certain height
  - Rolling
  - Pitching
  - Wheel load differences left/right
  - Axle load differences front/rear
- Variable vehicle longitudinal speed
- Vertical dynamics
- Tire modeling: mathematical/physical
- Used in professional vehicle dynamics simulation

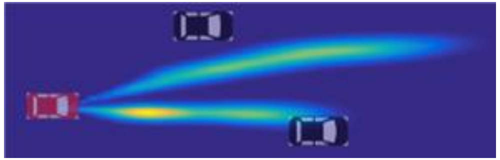


Vehicle Dynamics - Use Cases

- **Vehicle Dynamics Simulation** - Provide the correct behavior for your vehicle in simulation
- **State Estimation**: Calculate how your vehicle has moved
- **Behavior Prediction**: Calculate how other vehicles will move/behave
- **Model-Based Algorithms**:
  - Model Predictive Control (MPC)
  - Model based Reinforcement Learning







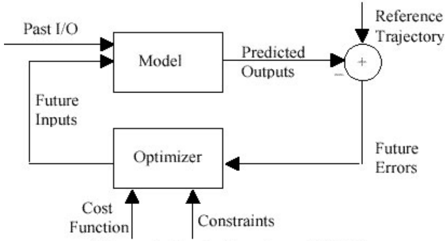


Figure 1: Basic Structure of MPC

Vehicle Dynamics Simulation

Vehicle Dynamics Simulation

- For prototype development, simple vehicle setups and testing, you can use the **CommonRoad Vehicle Dynamics Package (Python)** (Used in the F1Tenth Gym)

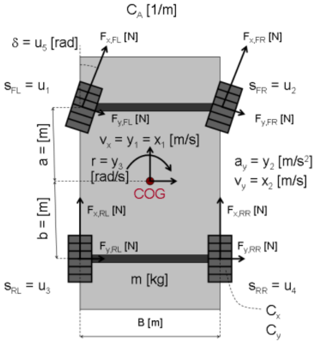
https://gitlab.lrz.de/tum-cps/commonroad-vehicle-models/blob/master/vehicleModels_commonRoad.pdf

- Different model types available
- Different parameters necessary (you need to identify them)
- Python package – Install with pip
- Explanation about the differential equations

Vehicle Dynamics Simulation

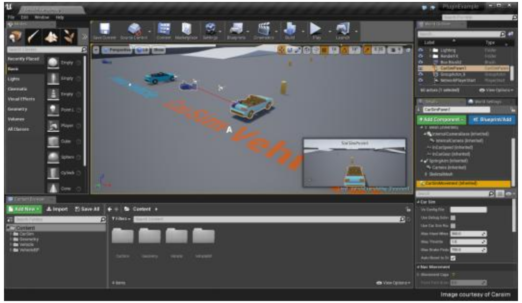
- The vehicle dynamics needs to be simulated/modeled, otherwise your car is not behaving properly.
- Some simulation environments come with Vehicle Dynamic Simulations:

Matlab

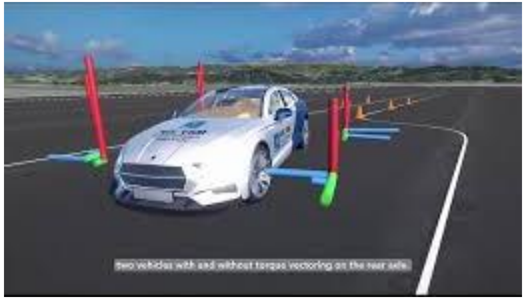


IPG CarMaker

Unreal/Unity /Carla

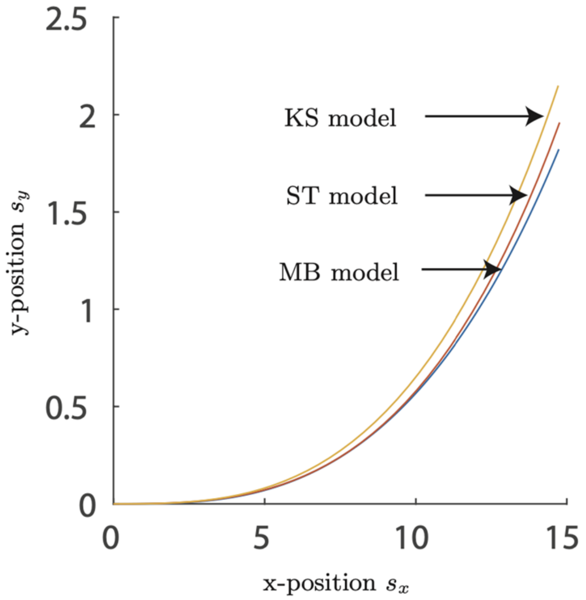


AVL VSM

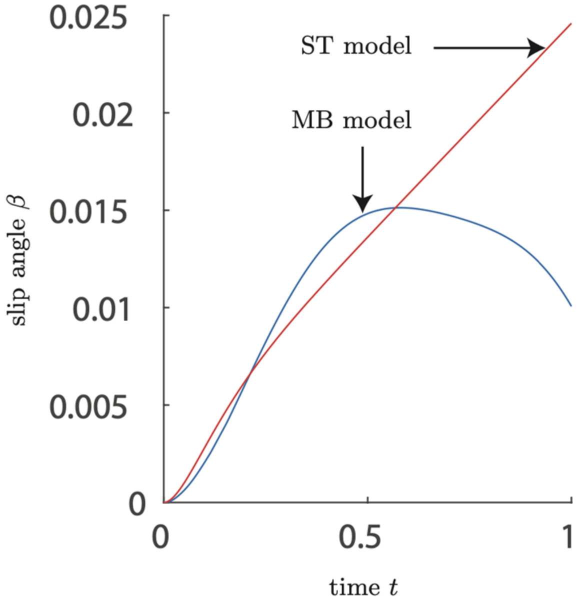


Example: Cornering

- Different model types lead to different behaviors



(a) Path of center of gravity.



(b) Slip angle.

Questions?